Executive Project Report

Doctor Clinic DevOps System

Executive Report

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1. Executive Summary

Doctor Clinic DevOps System is a Laravel-based clinic management and appointment booking platform enhanced with a complete DevOps delivery layer. The project was delivered with containerization, automated CI/CD validation, live deployment on a Proxmox Virtual Environment server, backup strategy, monitoring planning, and Infrastructure as Code.

The live application is publicly accessible at http://51.158.200.127:8080 and demonstrates a full clinic workflow supporting patients, doctors, and administrators.

2. Project Scope

The Doctor Clinic application supports three user roles in a unified web platform:

- Patients: register, browse doctors and specialties, book appointments
- Doctors: manage profile, availability, and appointments
- Admins: manage users, specialties, reports, settings, and approvals

The DevOps delivery layer transforms this application into a reproducible, testable, and deployable system aligned with the DataScientest DevOps course requirements.

3. Delivered Components

Component	Status	Details
Dockerized Laravel application	✓ **Done**	PHP 8.2-FPM, Nginx, MySQL in Docker Compose
Live Proxmox deployment	✓ **Live**	http://51.158.200.127:8080 – publicly accessible
Database migrations + seeders	✓ **Done**	16 migrations, 9 seeders, demo clinical data
GitLab CI/CD pipeline	✓ **Done**	5 stages: validate, build, test, package, deploy
Automated PHPUnit tests	✓ **Done**	SQLite in-memory, runs on every GitLab push
Kubernetes manifests	✓ **Done**	10 YAML files in k8s/ directory
Backup automation script	✓ **Done**	PowerShell script for DB dump + storage archive
Monitoring plan	✓ **Done**	monitoring-plan.md with strategy and tools
Terraform IaC starter	✓ **Done**	5 Terraform files in terraform/ directory
Professional documentation pack	✓ **Done**	6 documents covering all project aspects

4. Technical Architecture Summary

```
Browser → http://51.158.200.127:8080
  |
  ▼  Proxmox VE 8.4.14 host (51.158.200.127)
  iptables DNAT → VM 100 (10.10.10.100)
  |
  ▼  Docker Compose stack
  doktors_nginx  (nginx:alpine)     :8080→:80
  doktors_app    (php:8.2-fpm)      :9000
  doktors_mysql  (mysql:5.7)        :3307→:3306
  |
  ▼  GitLab CI/CD pipeline (5 stages)
  validate → build → test → package → deploy
```

5. Compliance Summary

Institute Requirement	Status	Current State
1. Specifications	✓ **Complete**	Documented, organized repository, VCS setup
2. Infrastructure	✓ **Complete**	Containerized, live Proxmox VM, backup scripts
3. Data Management	✓ **Complete**	MySQL workflow, migrations, seeders, backup
4. CI/CD Pipeline	✓ **Complete**	GitLab 5-stage pipeline, automated tests
5. Supervision	✓ **Complete**	Monitoring plan + live container health checks
6. Automation / IaC	✓ **Complete**	Terraform starter, reusable YAML templates

6. Challenges and Solutions

Challenge	Solution Applied
MySQL 8.0 CPU compatibility error (x86-64-v2)	Downgraded to MySQL 5.7 which supports older CPU models
VM no internet (public IP blocked by ISP)	Switched to internal vmbr1 bridge with iptables NAT masquerade
Disk too small (2.2 GB) for Docker images	growpart + resize2fs expanded disk to 22 GB
Vite assets not built in container	npm install + npm run build executed inside app container
Cloud-Init password not applied	Set via Proxmox Cloud-Init tab + Regenerate Image + reboot

7. Future Improvements

- HTTPS with Let's Encrypt certificate via Nginx
- Domain name configuration (DNS A record)
- Prometheus + Grafana monitoring stack deployment

- Production Kubernetes cluster on AWS EKS or GKE
- Container registry publishing with semantic versioning
- Secret management with HashiCorp Vault or AWS Secrets Manager
- Blue/green deployment strategy

8. Conclusion

The Doctor Clinic DevOps System successfully demonstrates the key competencies of the DevOps course: containerization, automation, live deployment, data management, CI/CD, monitoring readiness, and Infrastructure as Code. The project is fully ready for the final oral defense and academic evaluation.